GR 8

Interactive Hi-fi Figma Prototype

Company Name

Month Year

Logo

Meet Our Team

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Presentation Outline

1. Project Overview

Introduction to Thought AI and the problem the app is designed to solve.

2. Heuristic Evaluation Results

Summary of the GR7 evaluation including number of violations and severity levels.

3. Major Usability Issues Identified

Discussion of the most important severity level 3 and 4 violations found during evaluation.

4. Design Changes in the Hi-Fi Prototype

Explanation of the interface improvements made to address usability problems.

5. Impact of the Revisions

How the updated design improves usability and supports the goals of the application.

Problem Overview

People across many backgrounds increasingly view AI with skepticism. Many AI tools provide shortcuts that discourage critical thinking and replace human effort rather than supporting it. As companies adopt AI for speed and efficiency, tools often prioritize output over understanding. This has led to concern that human reasoning and problem solving are being devalued.

In education, this tension is especially visible. Students use AI to bypass reasoning, receive full solutions instantly, and disengage from the learning process. Our GR5 testing showed that while students want help, they do not want to lose control of their thinking. The challenge is designing AI that supports learning without replacing human reasoning.

Solution Overview

Instead of providing answers on demand, the AI withholds full solutions unless the student demonstrates active thinking.Progress is earned through visible reasoning, partial work, and engagement with guided prompts. The AI responds with hints, questions, and feedback rather than final answers, reinforcing learning without replacing human thought. This approach reframes AI as a support for reasoning rather than a shortcut.

Product Overview

Product: ThoughtAI

ThoughtAI is an iPad based learning app that applies this solution to handwritten problem solving.

Students can write directly on the screen or take photos of their work. ThoughtAI provides 3 forms of analyzing their reasoning and delivers contextual guidance, requiring students to complete each reasoning step before progressing or requesting further help.

By integrating AI into a student’s natural workflow, ThoughtAI supports reflection and understanding while preventing over reliance on AI generated answers.

Heuristic Evaluation Results

Heuristic Evaluation Results

- Based on the GR7 heuristic evaluations conducted by two independent evaluators, a large number of usability issues were identified across the ThoughtAI prototype. The evaluation revealed approximately **55 total heuristic violations** spanning multiple Nielsen usability heuristics.

- The violations ranged from **cosmetic design inconsistencies to major usability problems affecting navigation and user control**. A large portion of the issues were categorized as **severity level 3 and 4**, meaning they represented major usability problems that could significantly impact user interaction.

- The most frequently violated heuristics included:
  - User Control and Freedom
  - Help and Documentation
  - Aesthetic and Minimalist Design
  - Error Prevention

**Because these violations directly affect usability and task completion, the redesign of the high-fidelity prototype focused on addressing these major issues first.

Severity Breakdown

- The heuristic evaluation categorized usability issues into four severity levels based on their impact on user experience.

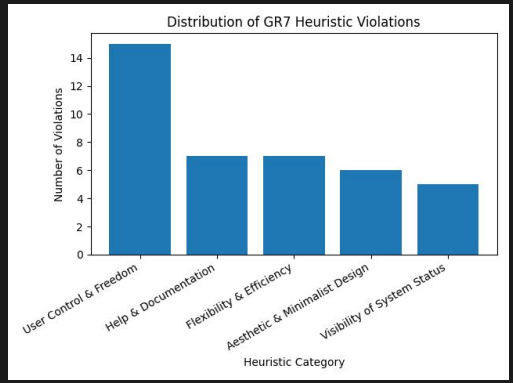
- **Severity 1:** Cosmetic issues that reduce visual polish but do not significantly impact usability.
- **Severity 2:** Minor usability issues that may create small inefficiencies or confusion.
- **Severity 3:** Major usability issues that significantly disrupt user workflows.
- **Severity 4:** Critical usability problems that prevent users from completing tasks or navigating the interface effectively.

- The evaluation results showed that **a large portion of violations fell into severity level 3**, particularly issues related to navigation control and system feedback.Because these violations affected core interactions within the application, they were prioritized during the redesign of the prototype.

Distribution of Heuristic Violations

- The heuristic evaluation revealed several patterns in the usability issues identified.

- The most frequently violated heuristics included:
  - **User Control and Freedom**
    Users were unable to exit certain workflows or undo actions.
  - **Help and Documentation**
    Several buttons and AI features lacked explanations for their functionality.
  - **Aesthetic and Minimalist Design**
    Some interface elements were misaligned or visually inconsistent.
  - **Error Prevention**
    Certain workflows could cause users to lose progress or perform unintended actions.



- These trends indicated that the interface required improvements in **navigation flexibility, feature clarity, and visual consistency**.

Major Severity 3 and 4 Violations Identified

- Several **severity 3 and severity 4 violations** were identified that directly impacted the usability of the prototype.
- One major issue involved **lack of navigation control**. Users could not exit certain workflows once they entered them. For example, when users clicked **New Note**, there was no option to return to the previous screen.
- Similarly, when users opened the **search interface**, there was no way to navigate back to the home page.
- Another major issue occurred in the **camera workflow**. Once users opened the camera interface, they were forced to take a photo before returning to the notebook.
- Users also **could not retake a photo** if the image was captured incorrectly.
- Additionally, once users activated **Tutor Mode**, there was **no option to exit the mode**, limiting user control.
- These issues created friction in the interaction flow and prevented users from easily correcting mistakes.

Remaining Severity 3 and 4 Violations

- Some severity level 3 usability issues were identified during the heuristic evaluation that were not fully addressed in the current hi-fi prototype due to design constraints within the Figma prototype.
- Examples include:

  · Users cannot paste text from their clipboard directly into the notebook interface:

  · Once a user creates a new note, there is currently no option to delete the note:

  · When importing a question through the camera feature, previous work may be deleted.

  · There is no confirmation screen when exiting the notebook during Tutor Mode.

  · The notebook does not currently include a save feature, meaning work may be lost when the user exits the notebook.

- These issues were documented as part of the heuristic evaluation but were outside the scope of the current prototype revisions. Future iterations of the design would address these problems to further improve user control, data persistence, and error prevention.

Severity 3 Violations Not Implemented in Prototype

Some severity level 3 usability issues identified during the heuristic evaluation were not implemented in the hi-fi prototype due to scope limitations and the focus on demonstrating core interaction workflows.

· **Users cannot paste text from their clipboard into the notebook interface**
The prototype focuses on the handwritten workflow that defines the core interaction of the app. ThoughtAI is designed around writing and camera input rather than typed responses. Adding clipboard functionality would introduce a different interaction pattern that was outside the scope of the prototype demonstration.

· **Once a user creates a new note, there is no option to delete it**
The prototype prioritized demonstrating note creation and the tutoring workflow rather than full notebook management features. Implementing deletion would require additional navigation screens and note management controls. These features would be added in a future iteration when expanding the notebook system.

· **Importing a question through the camera may delete previous work**
The prototype was designed to illustrate the camera capture interaction rather than complex data persistence behaviors. Supporting multiple states of saved work would require backend logic that is outside the scope of a Figma prototype. A real implementation would include safeguards such as confirmation dialogs or automatic saving.

· **There is no confirmation screen when exiting the notebook during Tutor Mode**
The prototype keeps the interaction flow simple in order to highlight the tutoring functionality itself. Adding confirmation dialogs would introduce additional interaction steps that were not necessary for demonstrating the core workflow. Future versions could include confirmation prompts to improve error prevention.

· **There is no save feature and work may be lost when exiting the notebook**
The prototype focuses on interaction design rather than full system functionality such as data persistence. Implementing a save system would require backend storage and session management beyond the scope of the prototype. In a real system, work would likely be automatically saved to prevent data loss.

UI Revisions Required Based On Evaluation

- Based on the heuristic evaluation results, several interface revisions were required to address the most critical usability issues.
- The redesign prioritized resolving all **severity 3 and severity 4 violations**, while also addressing several **severity 1 and severity 2 issues that could be easily improved**.
- Based on the heuristic evaluation results, the redesign required several major UI revisions to address many of the severity 3 and 4 usability violations identified in the evaluation.
- The following interface changes were implemented.
    - **Added back navigation for the search and camera interfaces**
      Added the ability to retake photos during the camera workflow
      Added an exit option for Tutor Mode
      Added an information page explaining the functionality of buttons and AI tools
      Adjusted the welcome page to navigate directly to the home page
      Centered the New Note text and corrected layout alignment
      Adjusted the pencil icon placement to improve visual consistency
- Additional improvements were made based on TA feedback.
    - **Improved button color palette to follow a more cohesive color scheme**
      Added padding to the OK button to improve spacing and clarity
- These revisions directly addressed the most severe usability violations identified during the evaluation.

How Evaluation Trends Guided Our Redesign

- The heuristic evaluation revealed that the most critical usability issues involved **navigation control and feature clarity**.
- As a result, the redesign prioritized improvements that would increase **user control and system transparency**.
- Navigation improvements ensured that users could easily exit workflows and undo actions.
- Feature documentation improvements ensured that users could understand the purpose of AI tools and system features.
- Visual design improvements addressed inconsistencies that reduced the overall clarity and polish of the interface.
- By addressing the **severity 3 and 4 violations first and resolving additional minor issues where possible**, the redesigned prototype provides a more intuitive and user-friendly experience.

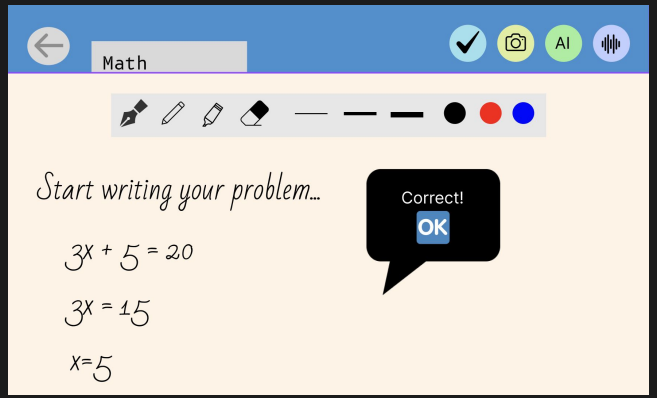
Connection Between Violations and Redesign

- The redesign of the prototype directly addressed the usability problems identified during the heuristic evaluation.
- Navigation violations were addressed by adding **back buttons and exit options** for workflows such as the camera interface and Tutor Mode.
- Camera workflow violations were addressed by introducing a **retake photo option**, allowing users to correct mistakes without restarting the process.
- Help and documentation violations were addressed by adding an **information page explaining the functionality of buttons and features**.
- Visual design violations were addressed by improving **button spacing, color consistency, and interface alignment**.
- These changes ensure that the updated prototype resolves the most critical usability problems identified in the evaluation.

Changes We Implemented

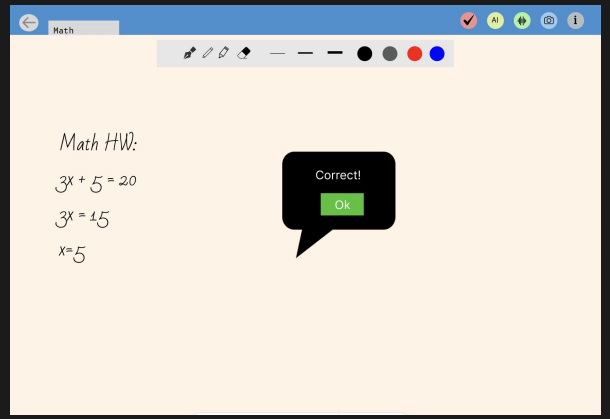
Design Changes in the Hi-Fi Prototype

Revision 1: Padded the ‘ok’ button



Before

A green box was added around the **“OK” button** to make the confirmation action more visually distinct within the feedback bubble. This improves visual clarity by helping users immediately recognize where to tap after receiving feedback. The change strengthens the visual hierarchy and makes the interaction more intuitive.



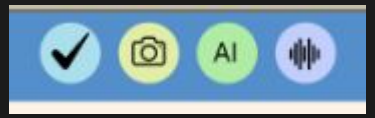
After

Revision 1

This revision was implemented to improve **visual clarity and action recognition** within the feedback interface. In the previous design, the “OK” button appeared small and visually blended into the feedback bubble, making it less obvious that it was the primary action users should take after receiving confirmation. By increasing the padding and highlighting the button with a green background, the updated design strengthens the **visual hierarchy** and clearly signals that the button is the next step in the interaction. This change improves **recognition rather than recall** by allowing users to immediately identify where to tap without searching the interface. As a result, the revised design better supports our usability goals of **reducing user hesitation and improving interaction efficiency**, while also creating a more consistent and intuitive feedback flow for completing tasks.

Design Changes in the Hi-Fi Prototype

Revision 2: Changed the buttons color theme for aestheticity



Before

The button colors were adjusted to create clearer visual distinction between different functions. The updated palette makes the primary action (check/confirm) more noticeable while keeping supporting tools visually organized. This improves both aesthetics and usability by helping users quickly identify and access key features.

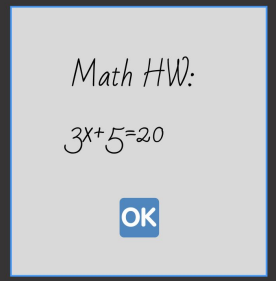
After

Revision 2

This revision was implemented to improve **visual consistency and functional clarity** within the interface. In the previous design, the button colors did not follow a cohesive palette, which made some features appear visually disconnected and made it harder for users to quickly distinguish between different actions. The updated design applies a more structured color scheme that clearly differentiates primary actions, such as confirming answers, from supporting tools like AI assistance and camera input. This strengthens the **visual hierarchy** of the interface and helps users quickly recognize the purpose of each button without needing to interpret icons alone. By improving the consistency of the color palette, the redesign supports our usability goals of **clearer feature recognition, faster interaction, and a more aesthetically coherent interface**, ultimately making the system easier and more intuitive for users to navigate.

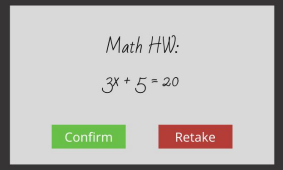
Design Changes in the Hi-Fi Prototype

Revision 3: Option to retake photo



Before

An option to retake the photo was added to give users more control before submitting their captured work. The previous design only allowed confirmation which could lead to frustration if the image was unclear or incorrect. Adding both Confirm and Retake options improves usability by allowing users to quickly correct mistakes and ensure better input quality.



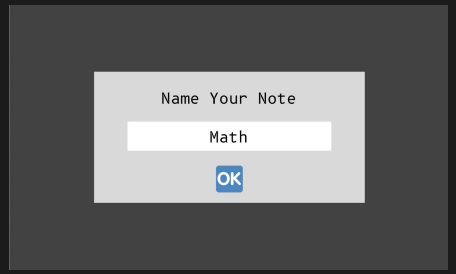
After

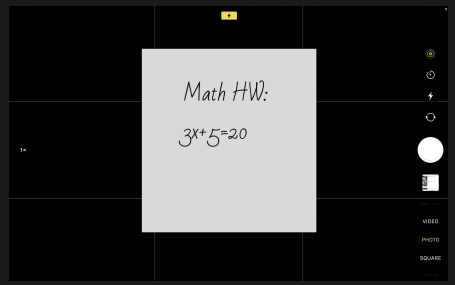
Revision 3

This revision was implemented to improve **user control and error recovery** within the camera workflow. In the previous design, users were required to confirm the captured image immediately, even if the photo was blurry or taken incorrectly. This limited the user's ability to correct mistakes and could create frustration if the image did not accurately capture the work. By adding a **Retake option alongside the Confirm action**, the updated design allows users to quickly correct errors before submitting their work. This change supports the usability principle of **User Control and Freedom** by allowing users to reverse an action without restarting the process. As a result, the redesign improves the overall interaction flow and helps ensure **more accurate submissions and a smoother task completion experience** for users.

Design Changes in the Hi-Fi Prototype

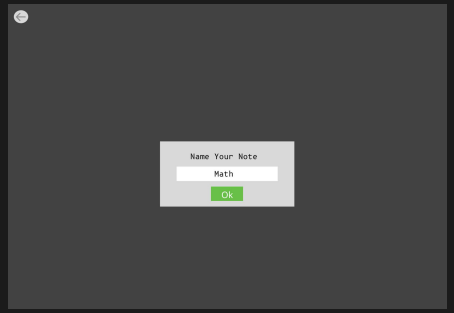
Revision 4: Added back button for search and camera

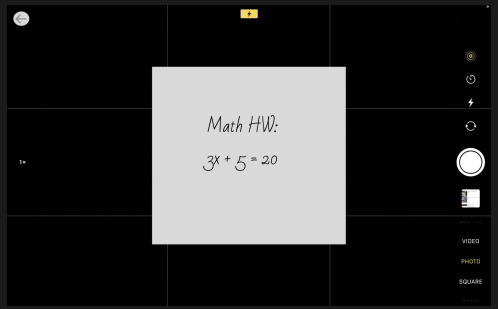




Before

A back button was added to the search and camera screens to improve navigation and give users a clear way to return to the previous interface. Without it, users could feel stuck or unsure how to exit these features. This change improves usability by providing a more intuitive and consistent navigation flow.





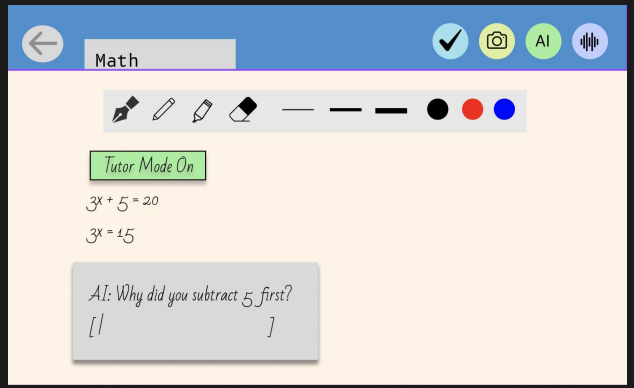
After

Revision 4

This revision was implemented to improve **navigation clarity and user control** within the interface. In the previous design, the search and camera screens did not include a clear way for users to return to the previous page, which could make users feel stuck or unsure how to exit the workflow. Adding a **back button** provides a familiar and intuitive navigation control that allows users to easily move between screens without interrupting their task. This change directly supports the usability principle of **User Control and Freedom**, ensuring that users can undo actions or leave a feature whenever needed. As a result, the redesigned interface creates a **more predictable and flexible navigation flow**, reducing confusion and helping users complete tasks more efficiently.

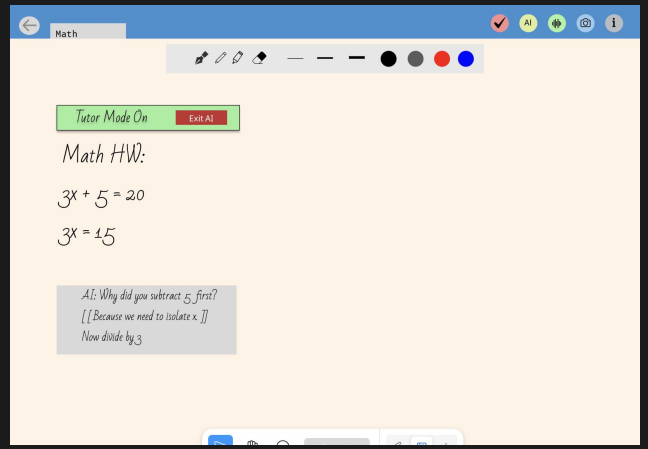
Design Changes in the Hi-Fi Prototype

Revision 5: Added an exit for the AI Tutor mode



Before

An exit button for AI Tutor mode was added so users can easily leave the tutoring interaction and return to normal note-taking. Without this option, users could feel stuck in the tutoring interface. This change improves user control and flexibility by allowing users to switch modes whenever they want.



After

Revision 5

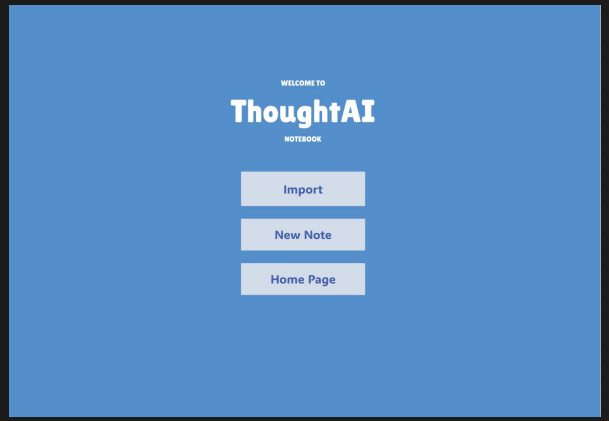
This revision was implemented to improve **user control and flexibility within the tutoring interaction**. In the previous design, once users entered AI Tutor mode, there was no clear way to exit the tutoring interface and return to normal note-taking. This could cause users to feel stuck in the tutoring state even if they no longer needed assistance. By adding an **Exit Tutor Mode button**, the updated design allows users to easily switch between tutoring and regular note-taking at any time. This change supports the usability principle of **User Control and Freedom** by ensuring that users can leave a feature without being forced to continue the interaction. As a result, the redesign creates a more flexible workflow and allows users to use AI assistance only when it is helpful, improving the overall usability and adaptability of the system.

Design Changes in the Hi-Fi Prototype

Revision 6: Added an option to go to home page from welcome page

Before

A Home Page option was added to the welcome screen to give users an additional navigation path within the app. Previously, users could only import or create a new note, which limited navigation choices. This change improves flexibility and makes it easier for users to access the main interface directly.



After

Revision 6

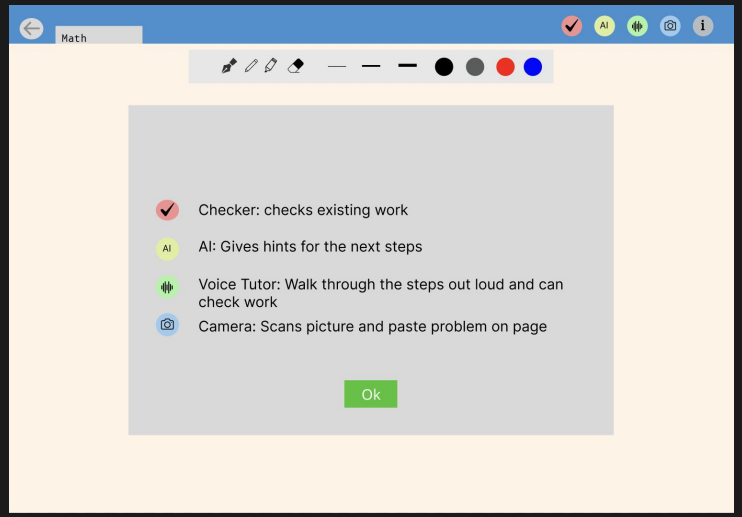
This revision was implemented to address a **User Control and Freedom** issue identified in the heuristic evaluation. In the earlier design, the welcome screen only allowed users to **import a problem or create a new note**, which limited navigation flexibility and made it harder for users to directly access previously created notebooks. Adding a **Home Page button** creates a clearer path into the main interface and reduces the friction caused by forcing users into only two actions at the start of the workflow. This change improves the usability goal of **efficient navigation** by giving users more control over how they enter the system and by better matching the different intentions users may have when opening the app. As a result, the revised design supports a more flexible and intuitive entry experience while aligning more closely with the heuristic evaluation results.

Design Changes in the Hi-Fi Prototype

Revision 7: Info page for all the buttons

Nothing Before

An info page explaining all the buttons was added to help users understand the purpose of each tool in the interface. Previously, users might not know what features like AI, voice tutor, or the checker do. This improves learnability by giving clear guidance on how each feature supports the user's workflow.



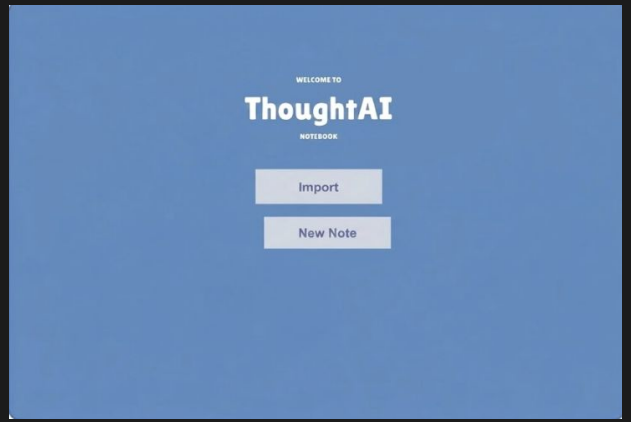
After

Revision 7

This revision was implemented to address a **Help and Documentation** issue identified in the heuristic evaluation. In the earlier design, several buttons such as the **AI helper, voice tutor, checker, and camera tools** did not include any explanation of their functionality, which could make it difficult for new users to understand how these features support their workflow. Adding an **information page that clearly describes the purpose of each button** improves the system's learnability and reduces the need for users to guess what each tool does. This change supports the usability goal of **recognition rather than recall**, allowing users to quickly understand the available features without needing prior experience with the interface. As a result, the updated design improves feature discoverability and helps users more confidently use the tools available within the application.

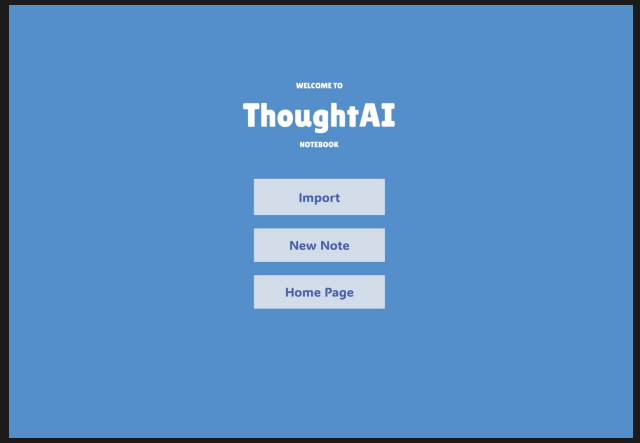
Design Changes in the Hi-Fi Prototype

Revision 8: Centered new note text



Before

The “New Note” text was centered to improve visual alignment and overall interface consistency on the welcome screen. In the previous design, the misaligned text created a slightly unbalanced layout that reduced visual clarity. Centering the text strengthens the visual hierarchy and creates a cleaner, more organized entry interface for users.



After

Revision 8

This revision was implemented to improve visual consistency and interface clarity on the welcome screen. In the previous design, the “New Note” text was slightly misaligned, which created a subtle imbalance in the layout and reduced the overall visual polish of the interface. Since the welcome page is the first screen users see, a clear and organized layout is important for guiding attention to the main actions.

By centering the “New Note” text and aligning it with the other elements, the updated design creates a more balanced and visually consistent layout. This change supports the Aesthetic and Minimalist Design heuristic by removing small visual inconsistencies and making the interface feel cleaner and easier to scan.

Progress towards each usability goal

Impact on Design Value

The revisions reinforce the core design value of supporting human reasoning rather than replacing it with AI output. ThoughtAI is designed to guide students through their thinking process using hints and feedback instead of giving full solutions, so maintaining user control and clarity in the interface is critical. By improving navigation flexibility, clarifying AI features, and strengthening visual hierarchy, the redesign ensures that students remain actively engaged in problem solving while using AI as a supportive tool. These changes help maintain the product’s goal of encouraging reasoning, reflection, and gradual progress rather than passive answer consumption.

Progress towards each usability goal

Progress toward usability goals

The revised designs significantly improved progress toward our usability goals of user control, feature clarity, and efficient interaction. Many of the most serious issues identified in the heuristic evaluation involved users being unable to exit workflows, correct mistakes, or understand the purpose of system features. The redesign addressed these problems by adding navigation controls (such as back buttons and exit options), improving error recovery through the retake photo feature, and increasing learnability through the information page explaining system tools. These changes directly respond to the major severity-3 usability violations identified in the evaluation and help users interact with the system more confidently and efficiently.

Progress towards each usability goal

Impact on project next steps and success metrics

The improvements also move the project closer to measurable success metrics related to task completion efficiency, reduced confusion, and improved feature discoverability. With clearer navigation and documentation, users should be able to move through workflows with fewer errors and less hesitation. However, some usability issues remain unresolved, such as the lack of save functionality, note deletion, and confirmation when exiting the notebook. Future iterations of the design will focus on addressing these remaining issues and further strengthening error prevention and data persistence to support a fully functional learning workflow.

Links

Figma:https://www.figma.com/design/cyCgAJYGy612MWPMXerbW9/ThoughtAI?node-id=0-1&p=f&t=o5a6k0vhD0Bp6Ld9-0

ReadME:https://docs.google.com/document/d/1VAVI8iyTiMWJcufleCyEmv0zfN8dpG8DJe1Fra4Yny0/edit?usp=sharing